

ADMM CASE STUDY

Smart City Traffic & Accident Analytics

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I. INTRODUCTION

1.1 Introduction:

Smart City Traffic & Accident Analytics is an analytics solution developed to analyze urban traffic patterns, congestion levels, accident characteristics, road-safety conditions, and accident-risk patterns.

The available datasets contain traffic sensor information and accident-related information such as location, date and time, vehicle count, average speed, congestion level, weather condition, road condition, vehicle type, accident severity, number of vehicles, casualties, and traffic density.

The objective of this case study is to transform the available traffic and accident datasets into a structured analytical solution that supports:

- Traffic volume analysis
- Traffic speed analysis
- Congestion analysis
- Congestion hotspot identification
- Time-based traffic analysis
- Peak and non-peak traffic analysis
- Accident analysis
- Accident severity analysis
- Accident-prone area analysis
- Weather-condition analysis
- Road-condition analysis
- Vehicle-type analysis
- Traffic-density analysis
- Accident-risk analysis
- Predictive risk exploration
- Interactive Power BI dashboards
- Traffic-management and road-safety insights

1.2 Business Problem & Objectives:

• Business Problem-

Urban traffic systems generate large amounts of traffic and accident-related data. However, raw datasets are difficult to analyze directly.

Without an integrated analytical solution, it is difficult to efficiently identify:

- Overall traffic volume
- Traffic patterns across locations
- Average traffic speed
- Congestion levels
- High-congestion locations
- Peak-hour traffic patterns
- Accident frequency
- Accident severity patterns
- Accident-prone locations
- Weather-related accident patterns
- Road-condition-related accident patterns
- Vehicle-type accident patterns
- Traffic-density accident patterns
- Time-based accident patterns
- Relative accident-risk patterns

The case study therefore focuses on building an analytical pipeline that converts the available traffic and accident datasets into structured business and road-safety insights.

• Objectives-

The key objectives of the project are:

- Extract traffic sensor and accident data from the available CSV datasets.
- Explore the datasets using Python and Pandas.
- Profile the datasets for record counts, data types, missing values, duplicates, and unique identifiers.
- Validate traffic and accident records.
- Validate date-time fields and categorical values.
- Clean and transform the source datasets.
- Perform feature engineering for date, time, peak-hour, congestion, and accident-severity analysis.
- Calculate the Congestion Index.
- Create traffic congestion categories for analytical comparison.
- Create Accident Severity Score.
- Build date, time, weather, road-condition, and vehicle-type dimensions.
- Build Fact_Traffic and Fact_Accident tables.
- Validate primary-key and foreign-key relationships.
- Load the transformed data into MySQL.
- Build a structured Data Warehouse using fact and dimension tables.
- Perform SQL-based traffic and accident analysis.
- Analyze traffic volume and speed by location and time.
- Analyze congestion patterns and high-congestion observations.

- Analyze accident frequency and severity.
- Analyze weather, road condition, vehicle type, and traffic density.
- Perform integrated traffic and accident analysis for the available overlapping traffic period.
- Develop an exploratory accident-risk model.
- Generate relative accident-risk scores and risk bands.
- Create an integrated Power BI data model.
- Develop DAX measures for traffic, congestion, accident, and road-safety KPIs.
- Develop interactive Power BI dashboards.
- Provide business insights based on observed traffic, congestion, accident, and risk patterns.

II. Architecture Overview

2.1 Architecture Overview

The **Smart City Traffic & Accident Analytics** solution follows a structured **ETL, Data Warehouse, Business Intelligence, and Predictive Analytics architecture**.

The process begins with the source traffic and accident datasets and progresses through data exploration, validation, cleaning, transformation, dimensional modeling, database loading, SQL analysis, predictive risk analysis, Power BI modeling, and dashboard development.

The major components of the architecture are:

1. Source Data

The project uses two CSV datasets:

- road_traffic_sensor_data.csv
- traffic_accident_data.csv

The traffic dataset contains traffic sensor observations, while the accident dataset contains accident, road, weather, vehicle, severity, casualty, and traffic-density information.

2. Python / Google Colab

Python and Pandas were used for:

- Data loading
- Data exploration
- Data profiling
- Missing-value analysis
- Duplicate analysis
- Primary-key validation
- Date-time validation
- Categorical-value validation
- Numerical analysis
- Data cleaning
- Data transformation
- Feature engineering
- Congestion analysis
- Accident analysis
- Predictive-model preparation
- Preparation of analytical datasets

3. ETL & Cleaned Data

The source datasets were cleaned, validated, transformed, and converted into dimensional-model tables.

Major processing activities included:

- Date-time conversion
- Data-type handling
- Duplicate validation
- ID validation
- Date validation
- Categorical validation
- Date feature creation
- Time feature creation
- Peak-hour classification
- Congestion Index calculation
- Congestion Category creation
- Accident Severity Score calculation
- Dimension-table creation
- Fact-table creation
- Primary-key validation
- Foreign-key validation

4. MySQL Data Warehouse

The cleaned data was loaded into the MySQL database:

smart_city_traffic_dw

The warehouse contains:

- Dim_Date
- Dim_Time
- Dim_Weather
- Dim_Road_Condition
- Dim_Vehicle_Type
- Fact_Traffic
- Fact_Accident

The fact tables store traffic and accident measurements, while the dimension tables provide time, weather, road-condition, and vehicle-type context.

5. SQL Business Analysis

SQL was used to perform:

- Traffic KPI analysis
- Traffic volume analysis
- Average-speed analysis
- Congestion analysis
- Location-wise traffic analysis
- Time-period analysis
- Peak-hour analysis
- Accident KPI analysis
- Accident severity analysis
- Location-wise accident analysis
- Weather analysis
- Road-condition analysis
- Vehicle-type analysis
- Traffic-density analysis
- Casualty analysis
- Integrated traffic-accident analysis

6. Predictive Risk Analysis

A logistic regression model was explored to classify accidents into a high-risk target based on accident characteristics.

The predictive process included:

- Defining Severe/Fatal accidents as the high-risk target
- Preparing model features
- Training the logistic regression model
- Evaluating model performance
- Generating risk scores
- Creating relative risk bands
- Analyzing risk by location, time, weather, road condition, vehicle type, and traffic density

The predictive results are treated as **exploratory relative-risk analysis**, rather than calibrated accident probabilities.

7. Power BI

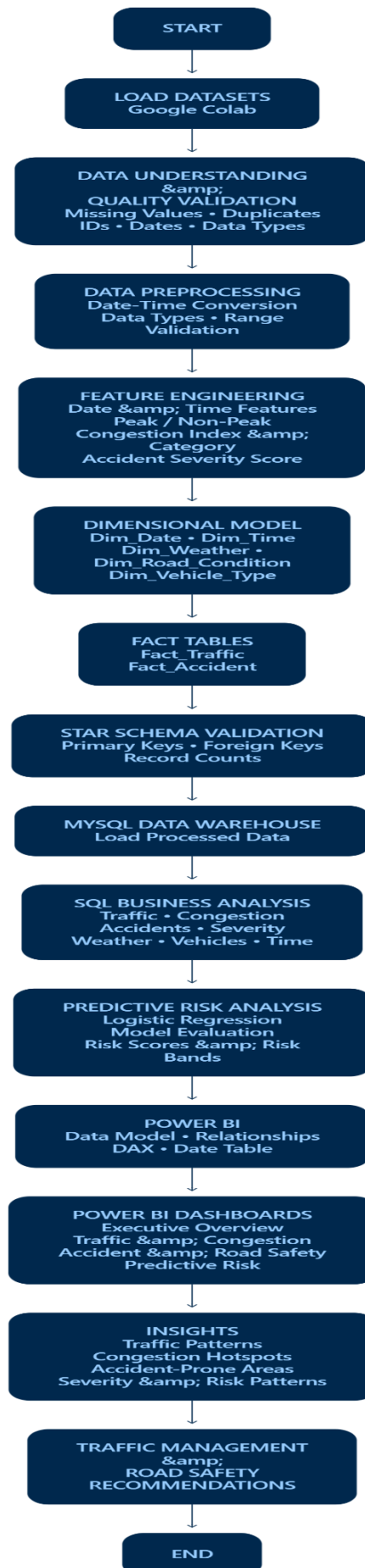
Power BI was used for:

- Data modeling
- Relationship creation
- DAX measures
- KPI development
- Interactive visualizations
- Slicers
- Traffic analysis
- Congestion analysis
- Accident analysis
- Road-safety analysis
- Predictive-risk analysis
- Insight reporting

The Power BI solution contains:

1. **Executive Overview**
2. **Traffic & Congestion Analysis**
3. **Accident & Road Safety Analysis**
4. **Predictive Risk Analysis**

2.2 Architecture Flow



III. PROJECT APPROACH

3.1 Methodology

Source Data Identification

Two source datasets were identified:

- Road Traffic Sensor Data
- Traffic Accident Data

The datasets were reviewed to understand their structure, fields, identifiers, dates, locations, and business meaning.

Initial Dataset Size

Road Traffic Sensor Dataset

- 300 records
- 6 original columns
- 300 unique Sensor IDs
- 5 locations

Traffic Accident Dataset

- 5,000 records
- 10 original columns
- 5,000 unique Accident IDs
- 5 locations

Data Extraction & Exploration

The datasets were loaded into Google Colab using Python and Pandas.

The datasets were profiled using:

- Record counts
- Column names
- Data types
- Missing-value analysis
- Duplicate analysis
- Unique ID analysis
- Numerical statistics
- Categorical analysis
- Date-range analysis
- Date-time analysis
- Traffic analysis

- Accident analysis

Data Validation

Validation checks were performed for:

- Missing values
- Duplicate records
- Unique Sensor IDs
- Unique Accident IDs
- Date-time fields
- Numerical fields
- Categorical fields
- Date ranges
- Traffic values
- Accident values

Both source datasets were validated with **zero missing values and zero duplicate records**.

Data Transformation

Feature engineering was performed to create analytical fields.

The following fields were created:

- Date
- Year
- Month
- Month_Name
- Day
- Day_Name
- Week
- Quarter
- Hour
- Is_Weekend
- Time_Period
- Peak_Hour
- Congestion_Index
- Congestion_Category
- Accident_Severity_Score

Time Period Classification

Traffic and accident records were classified into:

- Night
- Morning

- Afternoon
- Evening
- Late Evening

Peak-hour classification was also created:

- Peak
- Non-Peak

Congestion Index

For traffic observations, the Congestion Index was calculated using:

Congestion Index = Vehicle Count / Average Speed

This provides a derived measure for comparing traffic volume relative to observed speed.

Accident Severity Score

Accident severity was converted into an analytical numerical score:

- Minor = 1
- Moderate = 2
- Severe = 3
- Fatal = 4

This enabled numerical severity analysis.

Data Staging & Loading

The transformed datasets were exported as CSV files and prepared for MySQL loading.

The dimensional-model files included:

- Dim_Date.csv
- Dim_Time.csv
- Dim_Weather.csv
- Dim_Road_Condition.csv
- Dim_Vehicle_Type.csv
- Fact_Traffic.csv
- Fact_Accident.csv

MySQL Data Warehouse

The transformed tables were loaded into:

smart_city_traffic_dw

The warehouse follows a fact-and-dimension structure.

Power BI Data Modeling & Reporting

The MySQL Data Warehouse was connected to Power BI.

The following relationships were established:

- Dim_Date[Date_Key] → Fact_Traffic[Date_Key]
- Dim_Date[Date_Key] → Fact_Accident[Date_Key]
- Dim_Time[Time_Key] → Fact_Traffic[Time_Key]
- Dim_Time[Time_Key] → Fact_Accident[Time_Key]
- Dim_Weather[Weather_Key] → Fact_Accident[Weather_Key]
- Dim_Road_Condition[Road_Condition_Key] → Fact_Accident[Road_Condition_Key]
- Dim_Vehicle_Type[Vehicle_Type_Key] → Fact_Accident[Vehicle_Type_Key]

The model follows a **fact-and-dimension / star-schema structure**.

3.2 Technology Stack

Component	Technology	Purpose
Source Data	CSV	Traffic and accident data
Data Processing	Python, Pandas	Exploration, cleaning and transformation
Development Environment	Google Colab	Python-based preprocessing and analysis
Data Staging	CSV	Transformed analytical tables
Data Warehouse	MySQL	Structured analytical storage
Database Tool	MySQL Workbench	Database creation, ETL and SQL analysis
Business Analysis	SQL	Traffic, congestion and accident analysis
Predictive Analytics	Python / Scikit-learn	Exploratory accident-risk modeling
Visualization	Power BI	Interactive dashboards
Calculations	DAX	KPI and analytical measures
Reporting	Power BI Desktop	Traffic and accident reporting
Documentation	Microsoft Word	Project documentation

IV. SOURCE DATA & TABLE MAPPING

4.1 Source Data

The project uses two primary source datasets.

- **Traffic Sensor Dataset :** road_traffic_sensor_data.csv

Records: 300

- **Traffic Accident Dataset :** traffic_accident_data.csv

Records: 5,000

```
... Traffic Sensor Dataset:
Shape: (300, 6)

Accident Dataset:
Shape: (5000, 10)
```

4.2 Data Warehouse Table Mapping

Table	Type	Purpose
Dim_Date	Dimension	Date-related analysis
Dim_Time	Dimension	Hour, time period and peak analysis
Dim_Weather	Dimension	Weather-condition analysis
Dim_Road_Condition	Dimension	Road-condition analysis
Dim_Vehicle_Type	Dimension	Vehicle-type analysis
Fact_Traffic	Fact	Traffic sensor measurements
Fact_Accident	Fact	Accident measurements and attributes

V. DATA EXPLORATION & PRIMARY KEY VALIDATION

5.1 Data Exploration

Initial data profiling was performed using Python and Pandas.

The analysis included:

- Record counts
- Column names
- Data types
- Missing-value analysis
- Duplicate analysis
- Unique identifier analysis
- Numerical statistics
- Categorical analysis
- Date-time analysis
- Traffic analysis
- Accident analysis

Dataset	Records
Traffic Sensor Data	300
Traffic Accident Data	5,000
Dataset	Records

5.2 Employee ID Validation

The traffic dataset contained:

- 300 rows
- 300 unique Sensor IDs
- 0 missing values
- 0 duplicate records
- 5 locations

The traffic observations covered the period:

1 January 2024 – 13 January 2024

5.3 Attrition Data Validation

The accident dataset contained:

- 5,000 rows
- 5,000 unique Accident IDs
- 0 missing values
- 0 duplicate records
- 5 locations

The accident observations covered:

1 January 2024 – 27 July 2024

5.4 Feature Engineering

The following analytical features were created:

Date Features

- Date
- Year
- Month
- Month Name
- Day
- Day Name
- Week
- Quarter
- Weekend indicator

Time Features

- Hour
- Time Period
- Peak Hour

Traffic Features

- Congestion Index
- Congestion Category

Accident Features

- Accident Severity Score

VI. MYSQL DATA WAREHOUSE IMPLEMENTATION & DATA LOADING

6.1 MySQL DataBase Implementation

After data cleaning and transformation, the analytical tables were loaded into:

smart_city_traffic_dw

MySQL was used to provide structured analytical storage for traffic and accident facts and their supporting dimensions.



Figure 6.1: Tables created in MySQL Workbench

6.2 Data Warehouse Structure

- **Dim_Date**

Date_Key	Full_Date	Day	Day_Name	Week	Month	Month_Name	Quarter	Year	Is_Weekend
20240101	2024-01-01	1	Monday	1	1	January	1	2024	0
20240102	2024-01-02	2	Tuesday	1	1	January	1	2024	0
20240103	2024-01-03	3	Wednesday	1	1	January	1	2024	0
20240104	2024-01-04	4	Thursday	1	1	January	1	2024	0

- **Dim_Time**

	Time_Key	Time	Hour	Hour_Label	Time_Period	Peak_Hour
▶	0	00:00:00	0	00:00	Night	Non-Peak
	1	01:00:00	1	01:00	Night	Non-Peak
	2	02:00:00	2	02:00	Night	Non-Peak
	3	03:00:00	3	03:00	Night	Non-Peak
	4	04:00:00	4	04:00	Night	Non-Peak

- Dim_Weather

	Weather_Key	Weather_Condition
▶	1	Clear
	2	Fog
	3	Rain
	4	Snow
	5	Storm

- Dim_Road_Condition

	Road_Condition_Key	Road_Condition
▶	1	Dry
	2	Icy
	3	Under Construction
	4	Wet

- Dim_Vehicle_Type

	Vehicle_Type_Key	Vehicle_Type
▶	1	Bicycle
	2	Bus
	3	Car
	4	Motorcycle
	5	Truck

- Fact_Traffic

	Traffic_Key	Date_Key	Time_Key	Sensor_ID	Location	Vehicle_Count	Average_Speed	Congestion_Level	Congestion_Index	Congestion_Category	Date_Time
▶	1	20240101	0	a36f14	Downtown	90	59.00	High	1.5254	Low	2024-01-01 00:00:00
	2	20240101	1	256a39	Downtown	324	73.00	Moderate	4.4384	Moderate	2024-01-01 01:00:00
	3	20240101	2	a8a01f	Highway	61	56.00	Moderate	1.0893	Low	2024-01-01 02:00:00
	4	20240101	3	3036e3	Highway	90	56.00	High	1.6071	Low	2024-01-01 03:00:00
	5	20240101	4	2b45b1	Residential Zone	72	77.00	High	0.9351	Low	2024-01-01 04:00:00
	6	20240101	5	6204a2	Downtown	243	61.00	Moderate	3.9836	Low	2024-01-01 05:00:00
	7	20240101	6	10cf13	Residential Zone	353	74.00	Moderate	4.7703	Moderate	2024-01-01 06:00:00
	8	20240101	7	839bb8	Industrial Area	475	46.00	High	10.3261	High	2024-01-01 07:00:00

- Fact_Accident

	Accident_Key	Date_Key	Time_Key	Accident_ID	Location	Weather_Key	Road_Condition_Key	Vehicle_Type_Key	Accident_Severity	Accident_Severity_Score	Number_of_Vehicles	Casualties	Traffic_Density	Date_Time
▶	1	20240101	0	d02b3836	Suburbs	2	2	2	Minor	1	2	6	Low	2024-01-01 00
	2	20240101	1	694c8577	Downtown	1	2	2	Minor	1	2	2	Low	2024-01-01 01
	3	20240101	2	e0ed5c55	Residential Zone	4	1	4	Moderate	2	4	5	Moderate	2024-01-01 02
	4	20240101	3	6a78511a	Downtown	2	2	1	Moderate	2	3	8	Moderate	2024-01-01 03
	5	20240101	4	547b4c2	Highway	5	3	2	Moderate	2	4	1	High	2024-01-01 04
	6	20240101	5	1c7306c7	Suburbs	1	4	3	Fatal	4	2	8	Moderate	2024-01-01 05
	7	20240101	6	a5557a6b	Suburbs	5	2	4	Moderate	2	4	5	Moderate	2024-01-01 06
	8	20240101	7	2aef0a7	Downtown	2	4	2	Moderate	2	2	0	Low	2024-01-01 07

VII. POWER BI DATA MODEL & KPI DEVELOPMENT

7.1 Power BI Data Model

The MySQL Data Warehouse was connected to Power BI.

The model follows a fact-and-dimension structure with:

- Date relationships
- Time relationships
- Weather relationship
- Road-condition relationship
- Vehicle-type relationship

The dimension tables provide filtering and grouping context for the fact tables.

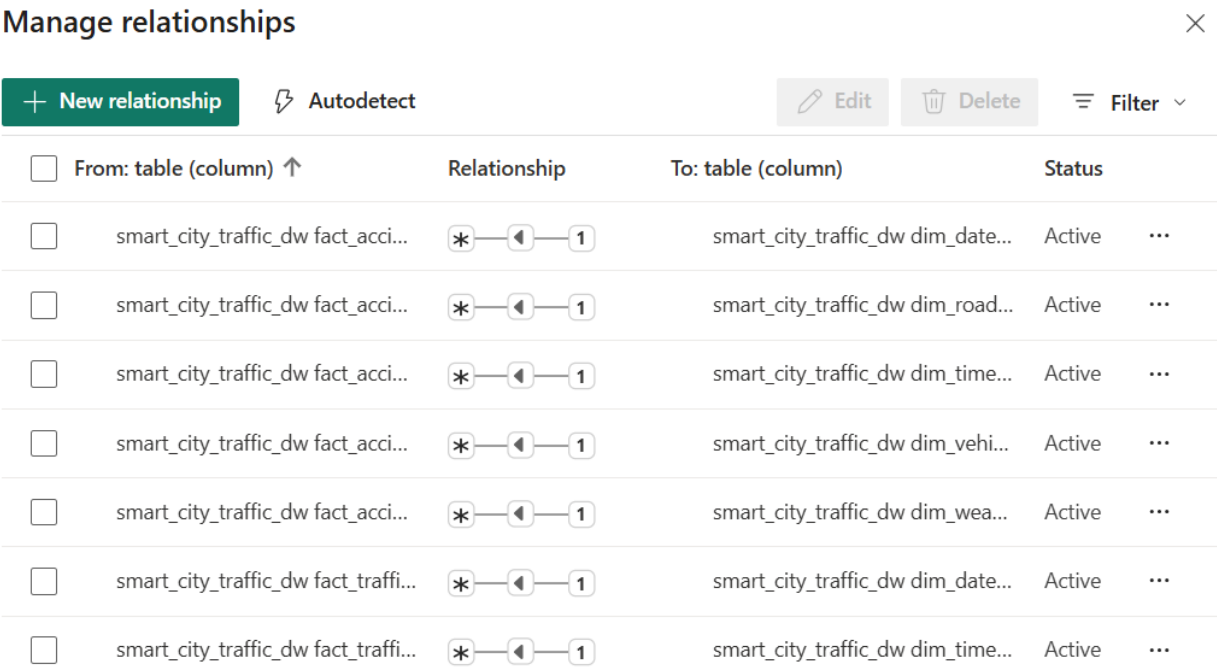


Figure 7.1: Power BI relational data model

7.2 DAX Measures & KPI Development

DAX measures were created in Power BI and organized in the dedicated:
The completed measures include:



Figure 7.2: DAX measures .

7.3 Power BI Data Model View

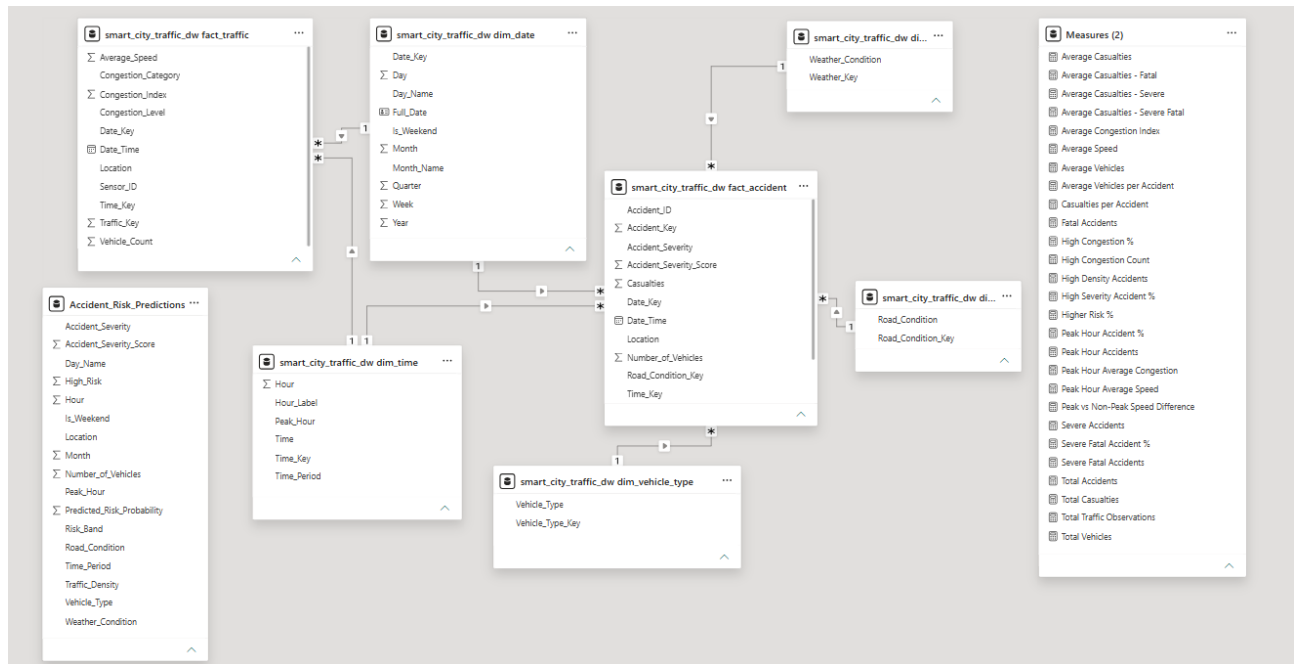


Figure 7.3: Employee Performance & Attrition Data Model

VIII. POWER BI DASHBOARDS & ANALYTICAL INSIGHTS

8.1 Executive Overview

The **Smart City Traffic & Accident Analytics** Executive Overview page provides a high-level view of traffic and accident activity.

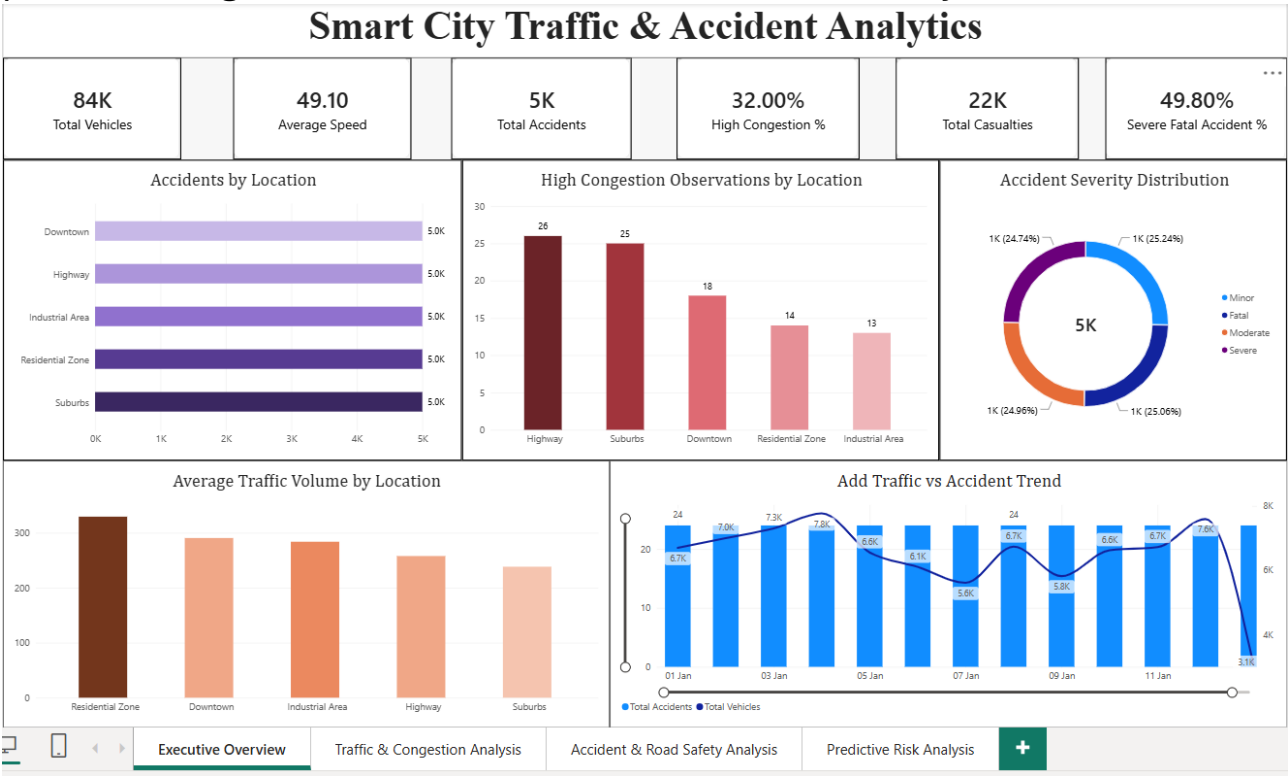


Figure 8.1: Executive Overview dashboard

8.2 Traffic & Congestion Analysis

The **Traffic & Congestion Analysis** page focuses on traffic behavior and congestion patterns.

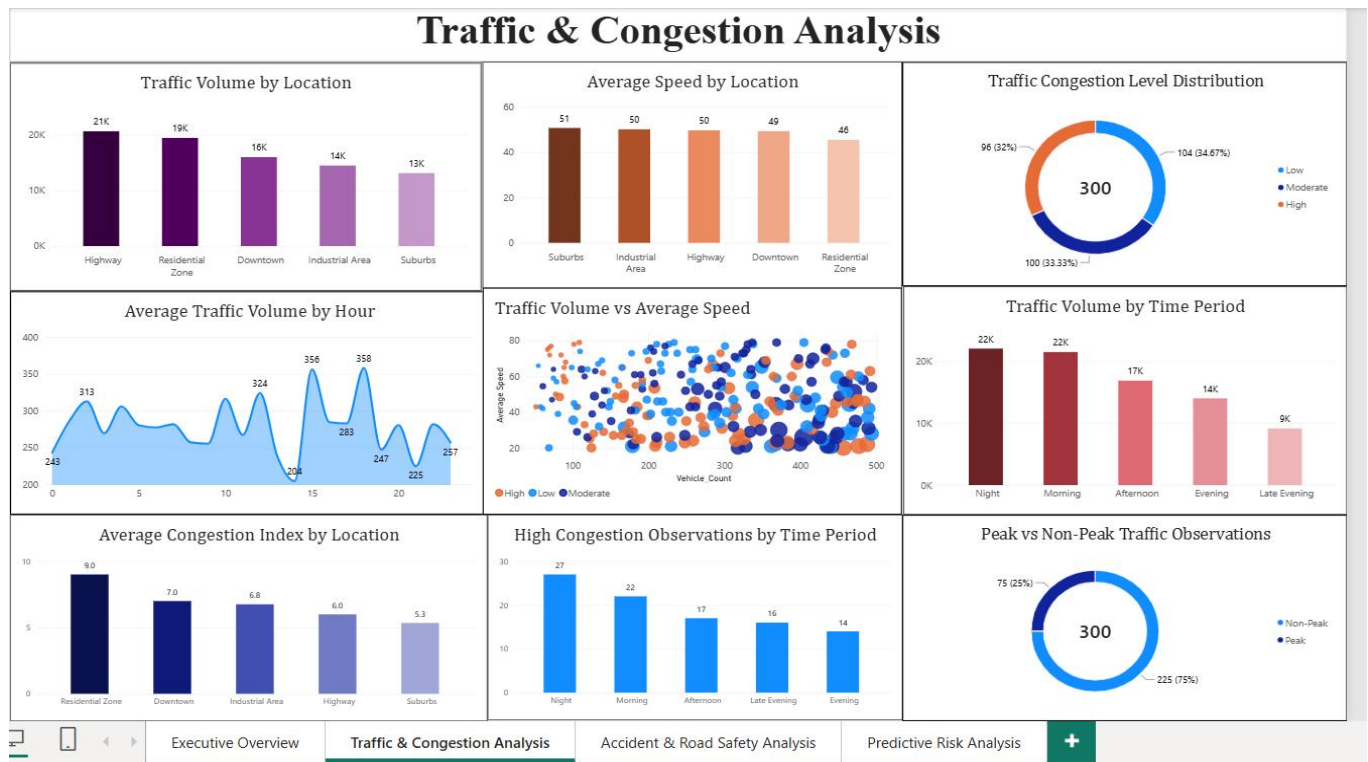


Figure 8.2: Traffic & Congestion Analysis dashboard

8.3 Accident & Road Safety Analysis

The **Accident & Road Safety Analysis** page focuses on accident patterns and road-safety characteristics.

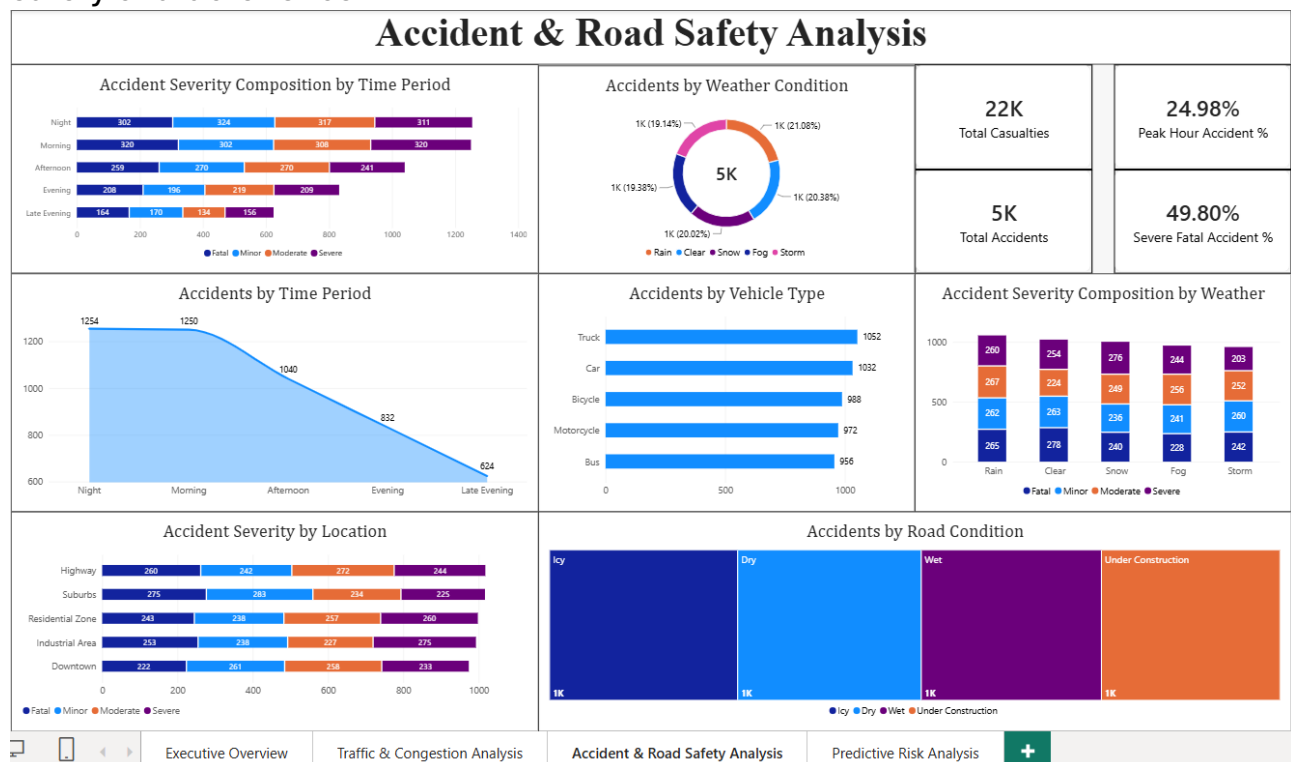
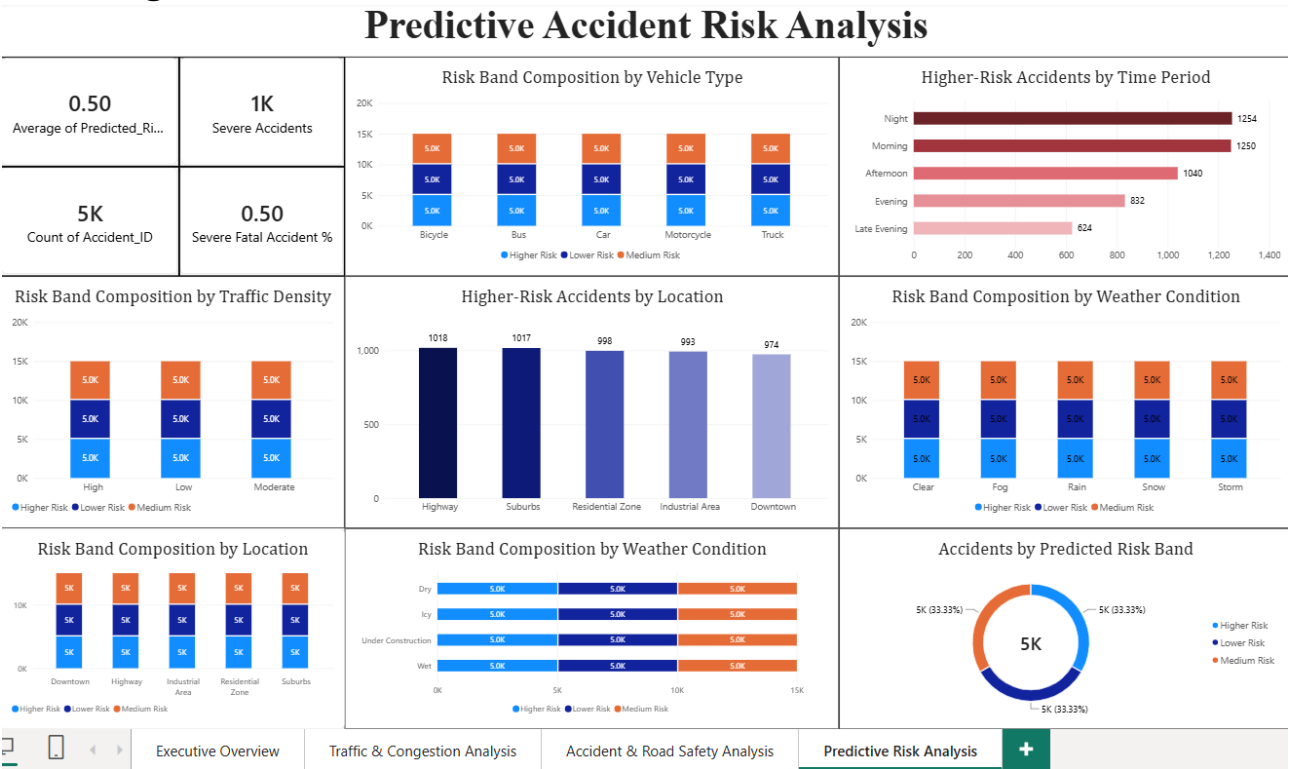


Figure 8.3 Accident & Road Safety Analysis Dashboard

8.4 Predictive Risk Analysis

The **Predictive Risk Analysis** page presents exploratory accident-risk modeling results.



IX. RESEARCH INSIGHTS

9.1 Traffic Insights

The traffic dataset contains:

- **300 traffic observations**
- **83,623 total vehicles**
- **278.74 average vehicles per observation**
- **49.10 average speed**
- **6.78 average congestion index**
- **96 high-congestion observations**

Among the locations, the observed traffic patterns differ in both traffic volume and congestion.

The Residential Zone has the highest average vehicle count at approximately **329.42 vehicles per observation**, while Suburbs has the lowest at approximately **238.36**.

The Residential Zone also records the highest average congestion index at approximately **9.00**.

9.2 Accident Insights

The accident dataset contains:

- **5,000 accidents**
- **22,467 casualties**
- **4.49 average casualties per accident**
- **1,253 fatal accidents**
- **1,237 severe accidents**
- **2,490 severe/fatal accidents**

Severe and fatal accidents together represent approximately **49.80%** of all recorded accidents.

9.3 Location-wise Accident Insights

The accident distribution across the five locations is relatively balanced.

Recorded accidents include:

- Downtown: 974
- Highway: 1,018
- Industrial: 993
- Residential: 998
- Suburbs: 1,017

This provides a basis for comparing accident characteristics rather than relying only on accident volume.

9.4 Time-Based Insights

Accidents were distributed across the defined time periods.

The recorded accident counts were:

- Night: 1,254
- Morning: 1,250
- Afternoon: 1,040
- Evening: 832
- Late Evening: 624

Peak-hour accidents accounted for **1,249 accidents**, approximately **24.98%** of the accident dataset.

9.5 Predictive Risk Insights

The exploratory logistic regression model produced relatively weak predictive discrimination, with a test AUC of approximately **0.551**. Therefore, the predictive analysis is used for **exploratory relative-risk segmentation**, rather than claiming reliable individual accident probabilities.

Relative Higher-risk patterns were examined across:

- Location
- Time period
- Weather
- Road condition
- Traffic density
- Vehicle type

X. CONCLUSION

11.1 Conclusion

The **Smart City Traffic & Accident Analytics** case study developed an end-to-end solution using **Python, MySQL, SQL, Power BI, and predictive modeling**. The project analyzed traffic, congestion, accidents, road-safety factors, and relative accident-risk patterns through structured data processing, data warehousing, SQL analysis, and interactive Power BI dashboards.

The solution provides a structured view of **traffic volume, congestion, accident severity, road conditions, and related risk patterns**. The developed dashboards help present these patterns clearly and support data-driven analysis of traffic and road-safety conditions.

11.2 Key Outcomes

- Implemented an end-to-end **traffic and accident analytics solution** using Python, MySQL, SQL, and Power BI.
- Created a structured **MySQL Data Warehouse** with fact and dimension tables.
- Analyzed **traffic volume, speed, congestion, accident severity, casualties, weather, road conditions, vehicle types, and time patterns**.
- Developed interactive Power BI dashboards for **executive overview, traffic analysis, accident analysis, and predictive risk analysis**.
- Applied predictive modeling to identify **relative accident-risk patterns and higher-risk groups**.
- Established a structured analytical framework to support **traffic management and road-safety analysis**.

11.2 Github Link

<https://github.com/surabhi-0405/CaseStudy-5-Smart-City-Traffic-Accident-Analytics>